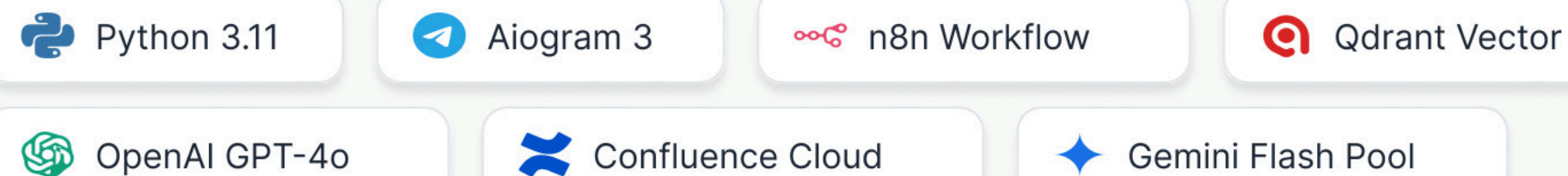


# AXON: Instant, Verified Answers from Complex Technical Wikis

Search hundreds of complex standards, schedule parameters, and UI workflows in 4 seconds.



97%

Composite Score

60+

Automated Tests

100+

Work Standards

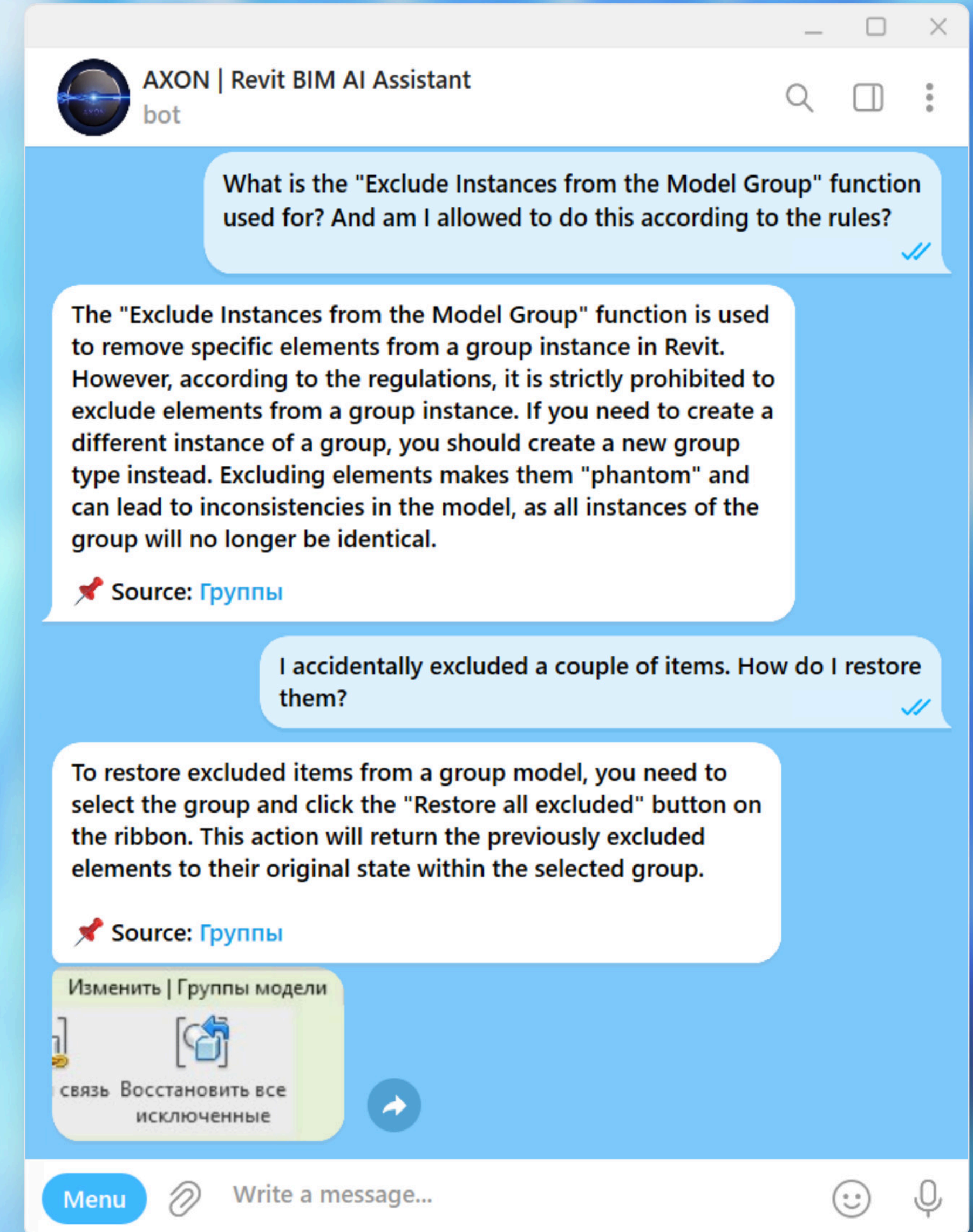
4.3s

Average Latency

Live 24/7 Production Deployment

Ubuntu VPS • 45MB RAM Footprint • Zero Downtime

Try Live: @bim\_axonbot ↗



# Information Chaos & The High Cost of Error in Technical Engineering

Standard corporate wikis cause massive engineer downtime, while generic LLMs hallucinate parameters and propose forbidden commands.

## 01 15+ Min Engineer Downtime / Query

Over 100 multi-page standards with 1,000+ UI screenshots, family parameters & visual SOPs scattered across Confluence.

TIME SINK

## 02 Tabular & Visual Context Fragmentation

Generic RAG chunkers sever parameter rows from table headers, completely ignoring family type dialogs, Revit UI settings, and animated SOPs prepared for specific modeling cases.

NAIVE RAG FAIL

## 03 High-Risk Modeling & On-Site Rework

Extrapolating forbidden commands (e.g. wall profile editing vs proper recess family) breaks schedules and leads to defective concrete formwork on site.

CRITICAL CLASH

! Critical Requirement: Technical AI assistance requires 100% verifiable citations and audit-proof compliance with corporate standards.

| AXON / Prohibited tools and actions                                                             |           |                                                                                                    |                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tool / Method                                                                                   | Status    | Engineering justification                                                                          |                                                                                                                                                        |
| Using the Composite Wall tool                                                                   | FORBIDDEN | It cannot be automatically reinforced with plugins or uploaded to estimates.                       | Model with independent parallel walls of the corresponding structural types.                                                                           |
| Editing a wall profile to create an opening or hole, or to adjust the height or width of a wall | LIMITED   | Breaks automatic reinforcement of wall ends by plugins.                                            | It can be used to create walls of varying heights for future reinforcement using the plugin. The wall height can be configured in the wall properties. |
| Attach top/base                                                                                 | FORBIDDEN | Creates uncontrolled geometric cuts when adjacent structures are modified.                         | Set dependency by linking to Levels.                                                                                                                   |
| Using tools Opening along the edge / vertical / in the wall , Shaft                             | FORBIDDEN | System openings do not contain parameters for opening lists and do not cut reinforcement.          | Use standard parametric families of holes KR.                                                                                                          |
| Using the Fencing Tool                                                                          | FORBIDDEN | Fences are implemented by the developers of the AR section; in the KR model they cause collisions. | In the KR section, monolithic parapets are modeled as walls.                                                                                           |
| Using the Models in Context tool without assigning the correct category                         | FORBIDDEN | Items without a category are not included in the bill of materials and cause coordination errors.  | It is necessary to assign a standard category to load-bearing structures.                                                                              |

# Hybrid Architecture: Precision Routing & Grounded Retrieval

Decoupled multi-tier pipeline combining a lightweight 24/7 Python daemon, visual graph orchestration, and millisecond vector retrieval.

T1

**Tier 1: Access & Client Gateway (Python / Aiogram 3)**  
2-Step FSM Access Funnel • Deep Link Lead Tracking (/start upwork)  
Silent Telemetry • Lightweight 24/7 Python Service (Docker & systemd ready)

GATEWAY

T2

**Tier 2: Pipeline Orchestration & Normalization (n8n Engine)**  
Domain Slang Normalization • Strict Intent Router  
Refusal Disjunction (Answer vs Grounded Out-of-Scope Isolation)

ORCHESTRATOR

T3

**Tier 3: Domain Vector Store & Index (Qdrant Vector DB)**  
text-embedding-3-small (Top-K=10) • Micro-chunk Merged Index (633 chunks)  
In-Memory HNSW Payload Filtering • Zero Table Split Guarantee

VECTOR DB

T4

**Tier 4: Generative Extraction & Grounding Guard (GPT-4o)**  
Strict Verbatim Attribute Extraction • Lex Specialis Priority  
Anti-Prompt-Patching Meta-Rules • Disjunction Grounding Guard

EXTRACTOR

Production Reliability: Deterministic daemon with PID lockfile & Docker containerization.

n8n Orchestrator • Production Workflow Graph

```
graph TD
    Webhook[Webhook] -- POST --> KR[KR Slang Normalizer]
    KR --> If[If]
    If -- true --> AI[AI Extractor]
    If -- false --> QR[Query Rewriter]
    QR --> AI
    QR --> GPT4oMiniRewriter[GPT-4o-mini Rewriter]
    QR --> SimpleMemory[Simple Memory]
    AI --> GPT4oMiniExtractor[GPT-4o-mini Extractor]
    AI --> Qdrant[Qdrant Vector Store]
    AI --> OpenAIEmbeddings[OpenAI Embeddings]
```

**Qdrant Vector Database • Collection: KR-BIM**  
OpenAI text-embedding-3-small • Isolated Self-Hosted Deployment

633  
Indexed Chunks  
Merged tables from Confluence  
ZERO TABLE SPLIT

Top-10  
Context Depth  
Dynamic payload window per query  
FILTERED PAYLOAD

1,536  
Dimensions  
Deep semantic vector embedding  
OPENAI SMALL

<50ms  
Lookup Speed  
Instant vector graph traversal  
SUB-SECOND

OPTIMIZED

# Overcoming Hard Limits of Standard RAG Systems

Four specialized pipeline solutions, engineered specifically for high-precision technical documentation and compliance.

01 Multi-Frame VLM Document OCR

Enrichment across 1,000+ attachments via VLM. Parameter table extraction & invisible Confluence metadata injection.

Accuracy: 55.6% → 96.4%

02 Cross-Lingual Semantic Bridge

English query → Russian standards retrieval → English response with verbatim Cyrillic parameter names in backticks.

87.5% Pass Rate • 100% Verbatim

03 Micro-Chunk Merging & Integrity

Zero split tables. Merging micro-sections (<150 chars) and sanitizing <td>/<th> cells without spurious subheadings.

0 Broken Property Tables

04 Strict Refusal Grounding

Terminal disjunction: immediate polite refusal for Out-of-Scope queries (SCAD/Robot) with full stop, zero dangling citations.

0 False Source Attributions

Zero Prompt-Patching Rule: Meta-architectural rules solve errors permanently across the domain without test-set overfitting.

AXON | Revit BIM AI Assistant

bot

In schedule "Спецификация сборных элементов", which fields compose the parameter "Расч\_Наименование"? ✓

The parameter "Расч\_Наименование" in the "Спецификация сборных элементов" is composed of two fields:  
1. ADSK\_Наименование  
2. ADSK\_Марка  
The field takes correct values only when both parameters are filled.  
Source: Спецификация сборных элементов

| Обозначение      | Наименование             | Кол. | Мас |
|------------------|--------------------------|------|-----|
| В                | С                        | Д    |     |
| Конструкции      |                          |      |     |
| /04-1-5-КР1 л.4  | Расч_Наименование        | К    |     |
| /04-1-5-КР1 л.9  | Основной блок Ш ЛП40с-29 | 2    |     |
| /04-1-5-КР1 л.14 | Основной блок Ш ЛП40с-28 | 60   |     |
| /04-1-5-КР1 л.16 | Верхний блок Ш ЛП40с-28  | 6    |     |
|                  | Верхний блок Ш ЛП40с-9   | 7    |     |

✓ VLM SEMANTIC MATCH VERIFIED

• Image 7442718.png dynamically delivered from Confluence Cloud attachments.  
• Parameter mask extracted from raster:  
ADSK\_Наименование + ADSK\_Марка

100% FACTUAL RESOLUTION (0 DOWNTIME)

Имя объедин. параметра:  
Расч\_Наименование

Объединенный параметр:

| Имя               | Префикс | Выборочное   | Суффикс | Разделите |
|-------------------|---------|--------------|---------|-----------|
| ADSK_Наименование |         | ADSK_Наимено |         |           |
| ADSK_Марка        |         | ADSK_Марка   |         |           |

Menu Write a message...

MULTI-FRAME VLM EXTRACTION → CONFLUENCE CONTENT PROPERTIES API PUT .../property/media\_captions

"7442718.png": "Configuring combined parameter «Расч\_Наименование» in precast schedule by merging ADSK\_Наименование and ADSK\_Марка. Merged parameters form a single string for the Name column in Revit precast schedule table."

ETL Pipeline: Confluence Content Properties → Injected into n8n Markdown Chunks → Indexed in Qdrant Vector Store

# Rigorous Multi-Criteria E2E Benchmarking

Automated regression testing across 60+ production scenarios evaluated by an asynchronous LLM-as-a-Judge pipeline.

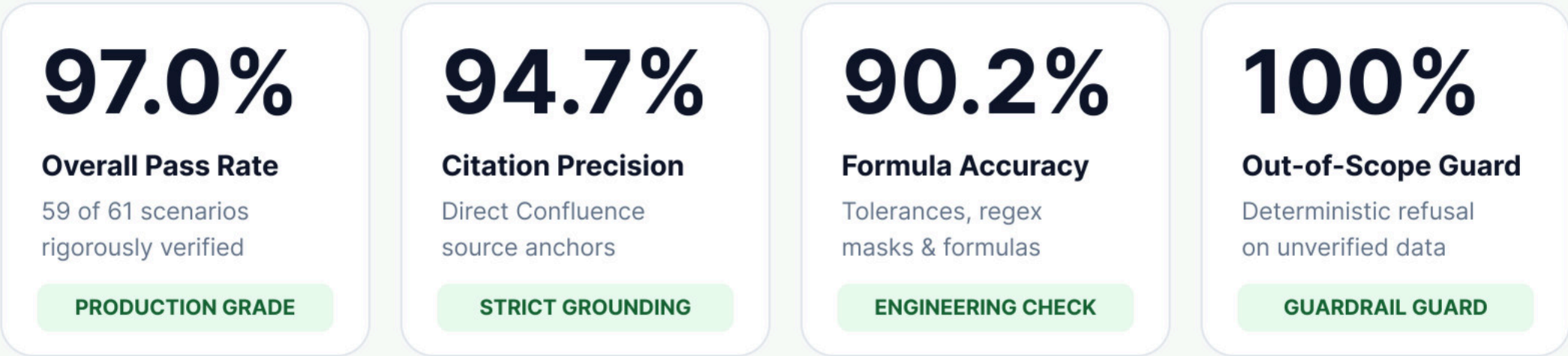
- ✓

**Factual & Engineering Precision**  
Validates Revit shared parameters, project attributes, regex masks, and embedment formulas.
- ✓

**Normative Prohibition Compliance**  
Prohibitions take absolute priority over generic commands. Zero false permits for edits.
- ✓

**Visual Context & Media Grounding**  
Verified extraction of Confluence detail schematics, interface workflows, and animated SOPs for specific modeling tasks.
- ✓

**Cross-Lingual Consistency**  
Full semantic parity and verbatim Cyrillic parameter preservation between English and Russian.



```
pytest Tests/test_e2e_rag_benchmark.py --suite=production

===== test session starts =====
platform linux -- Python 3.11.9, pytest-8.3.2, pluggy-1.5.0
evaluator: Asynchronous LLM-as-a-Judge (Gemini Flash Multi-Key Pool)
collected 61 items (45 stress cases + 16 cross-lingual benchmarks)

test_rebar_anchorage_and_lap_formulas PASSED [4.12s]
test_wall_profile_direct_editing_prohibition PASSED [3.85s]
test_confluence_visual_schematic_grounding PASSED [4.68s]
test_out_of_scope_structural_software_refusal PASSED [2.40s]
test_cross_lingual_ru_parameters_preservation PASSED [4.90s]
test_rebar_couplers_and_specialized_families PASSED [3.95s]

===== AXON BENCHMARK FINAL REPORT =====

Total Production Cases Evaluated      61 (45 Stress + 16 Cross-Lingual)
Execution Progress                    [=====] 100%
Benchmark Pass Rate (Composite)      97.0% [GRADE: EXCELLENT]
Retrieval & Grounding Accuracy        94.7% (Direct Confluence Links)
Engineering Precision (Revit)         90.2% (Attributes & Formulas)
Grounded Schematics & Media           100.0% (Verified Confluence Media)
Out-of-Scope Refusal Guard            100.0% (Strict Refusal Grounding)
Average Production Latency (P50) :    4.30s (Sub-second Qdrant Search)

-----
VERDICT: ALL CRITICAL STANDARDS VERIFIED [PRODUCTION GRADE]
```

BUSINESS IMPACT • PRODUCTION READY

# Measurable Business Value & Seamless Adoption

Seamless enterprise integration delivering concrete operational ROI by automating compliance without forcing workflow disruptions.

## 01 97% Search Latency Reduction: 15 min → 30 sec

30-60 DAYS PAYBACK

Cutting standard search from 15 minutes to 30 seconds eliminates costly context switching and frees senior BIM staff from repetitive support queries.

Minimal infrastructure TCO: sub-\$100/month API and self-hosted VPC footprint.

## 02 100% Verifiable Source Grounding

DIRECT WIKI CITATIONS

Every generated answer provides a direct 1-click citation to official Confluence pages with verified parameter tables. Prevents costly on-site errors and unauthorized modeling.

Automated workspace invitations delivered to authorized engineering leads.

## 03 Private Infrastructure & On-Premise Deployment

ZERO DATA LEAKAGE

Deploy on client's dedicated VPS, private cloud, or local on-premise servers. Zero external data exposure or proprietary model retraining.

Full audit logging and silent telemetry with zero operational overhead.



Ivan Kiriushov

Workflow Automation Architect | Python | n8n | AI Agents



### Telegram Assistant

Ubuntu VPS 24/7

Instant search across KR BIM standards with direct delivery of schematics & SOPs.

Try Live Bot in Telegram: @bim\_axonbot ↗



### Full System Video Demo

Voiceover Screencast • 2.5-MIN DEMO

End-to-end walkthrough of core engineering cases with cross-lingual RAG extraction.

Watch on YouTube ↗



### Architecture & Benchmark

Clean Architecture

Reproducible Docker container, systemd daemon, and 61-case LLM-as-a-Judge benchmark suite.

Explore GitHub Repository & Technical Specs ↗

Book a Live Architecture Review / Hire on Upwork ↗